

A c_0 counterexample to the gamma-multiplier strengthening

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Source problem

The source paper is Jan van Neerven, *γ -Radonifying operators – a survey*, arXiv:0911.3788 [1]. The question appears on PDF page 24, immediately after Theorem 5.2, the γ -bounded multiplier theorem.

It appears to be an open problem whether the operator $\widetilde{M}$ actually takes values in $\gamma(L^2(A; H), E)$ even in the simplest possible setting $A = (0, 1)$ and $H = \mathbb{R}$. Of course, an affirmative answer for Banach spaces E not containing an isomorphic copy of c_0 is obtained through an application of Theorem 4.2. □

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We continue with some examples of γ -bounded families. The first two results are due to WEIS [119].

Example 5.3. Let $(A, \mathcal{A}, \mu)$ be a σ -finite measure space and let $\mathcal{T}$ be a γ -bounded

Theorem 5.2 says that if $M : A \rightarrow \mathcal{L}(E, F)$ is strongly measurable with γ -bounded range, then

$$\widetilde{M} : \gamma(L^2(A; H), E) \longrightarrow \gamma_\infty(L^2(A; H), F)$$

is bounded. The paragraph after the proof asks whether $\widetilde{M}$ actually takes values in the radonifying class γ , already for $A = (0, 1)$ and $H = \mathbb{R}$. The displayed paragraph writes the codomain as $\gamma(L^2(A; H), E)$; in the theorem's notation the evident codomain is $\gamma(L^2(A; H), F)$. The counterexample below works even in the same-space case $E = F = c_0$.

Definitions used

Let H be a Hilbert space and E a Banach space. An operator $T : H \rightarrow E$ is γ -summing if

$$\|T\|_{\gamma_\infty(H, E)}^2 = \sup_{(h_k)} \mathbb{E} \left\| \sum_{k=1}^N \gamma_k T h_k \right\|_E^2 < \infty,$$

where the supremum is over all finite orthonormal systems in H . It is γ -radonifying if the corresponding Gaussian series converges in $L^2(\Omega; E)$ for an orthonormal basis; equivalently $\gamma(H, E)$ is the closed radonifying subspace of $\gamma_\infty(H, E)$.

A family $\mathcal{T} \subseteq \mathcal{L}(E, F)$ is γ -bounded if there is C such that for all finite choices $T_j \in \mathcal{T}$ and $x_j \in E$,

$$\left(\mathbb{E} \left\| \sum_j \gamma_j T_j x_j \right\|_F^2 \right)^{1/2} \leq C \left(\mathbb{E} \left\| \sum_j \gamma_j x_j \right\|_E^2 \right)^{1/2}.$$

Proof Intuition

The source survey already recalls the Linde–Pietsch diagonal operator

$$D : \ell_2 \rightarrow c_0, \quad D(\alpha_n) = (\alpha_n / \sqrt{\log(n+1)})_n,$$

which is γ -summing but not γ -radonifying. The point is to realize exactly this diagonal operator as $T_{M\phi}$ for a multiplier whose range is still γ -bounded. Rademacher functions provide the orthonormal coordinates in $L^2(0, 1)$, while the slow weights $(\log(n+1))^{-1/2}$ are just small enough for all finite Gaussian maxima to be uniformly square-integrable and just large enough that the full Gaussian coordinate sequence fails to tend to zero.

Lemma (A Gaussian maximal estimate). *Let $a_n = (\log(n+1))^{-1/2}$. There is an absolute constant C such that, for every finite or countable centred real Gaussian family (Y_n) satisfying $\text{Var}(Y_n) \leq \sigma^2$ for all n ,*

$$\mathbb{E} \sup_n |a_n Y_n|^2 \leq C^2 \sigma^2.$$

Proof. It suffices first to bound finite suprema; the countable case then follows by monotone convergence. By scaling assume $\sigma = 1$. For $u \geq 3$, the Gaussian tail estimate $\mathbb{P}\{|Y_n| > v\} \leq 2e^{-v^2/2}$ and the union bound give

$$\mathbb{P}\{\sup_n |a_n Y_n| > u\} \leq 2 \sum_{n \geq 1} \exp\{-u^2 \log(n+1)/2\} = 2 \sum_{n \geq 1} (n+1)^{-u^2/2}.$$

Since $u \geq 3$,

$$\sum_{n \geq 1} (n+1)^{-u^2/2} \leq 2^{-(u^2/2-2)} \sum_{n \geq 1} (n+1)^{-2}.$$

Therefore

$$\mathbb{E} \sup_n |a_n Y_n|^2 = \int_0^\infty 2u \mathbb{P}\{\sup_n |a_n Y_n| > u\} du \leq 9 + C_0 \int_3^\infty u 2^{-u^2/2} du < \infty,$$

with an absolute constant independent of the finite family. $\square$

Counterexample. *The multiplier theorem in [1, Theorem 5.2] cannot in general be strengthened from γ_∞ -valued to γ -valued, even for $A = (0, 1)$, $H = \mathbb{R}$, and $E = F = c_0$.*

Proof. Let $(r_n)_{n \geq 1}$ be the Rademacher functions on $(0, 1)$, viewed as an orthonormal sequence in $L^2(0, 1)$, and put

$$a_n = \frac{1}{\sqrt{\log(n+1)}}, \quad g(t) = (a_n r_n(t))_{n \geq 1} \in c_0.$$

The map $g : (0, 1) \rightarrow c_0$ is strongly measurable, since it is the uniform limit of its finite-coordinate truncations.

Let $e_1^* : c_0 \rightarrow \mathbb{R}$ be the first coordinate functional and define

$$M(t)x = e_1^*(x)g(t), \quad x \in c_0.$$

Then $M : (0, 1) \rightarrow \mathcal{L}(c_0, c_0)$ is strongly measurable. Its range is γ -bounded. Indeed, for finite $t_j \in (0, 1)$ and $x_j \in c_0$, set $b_j = e_1^*(x_j)$. For each coordinate n , the Gaussian variable $\sum_j \gamma_j b_j r_n(t_j)$ has variance at most $\sum_j |b_j|^2$. Lemma gives

$$\mathbb{E} \left\| \sum_j \gamma_j M(t_j) x_j \right\|_{c_0}^2 = \mathbb{E} \sup_n \left| a_n \sum_j \gamma_j b_j r_n(t_j) \right|^2 \leq C^2 \sum_j |b_j|^2.$$

Since e_1^* is contractive,

$$\sum_j |b_j|^2 = \mathbb{E} \left| \sum_j \gamma_j e_1^*(x_j) \right|^2 \leq \mathbb{E} \left\| \sum_j \gamma_j x_j \right\|_{c_0}^2.$$

Thus $\{M(t) : 0 < t < 1\}$ is γ -bounded.

Now take the constant finite-rank simple function

$$\phi(t) : \mathbb{R} \rightarrow c_0, \quad \phi(t)\lambda = \lambda e_1.$$

The induced operator $T_{M\phi} : L^2(0, 1) \rightarrow c_0$ is

$$T_{M\phi}f = \int_0^1 f(t)g(t) dt = (a_n \langle f, r_n \rangle_{L^2})_{n \geq 1}.$$

Bessel's inequality implies $(\langle f, r_n \rangle)_n \in \ell_2$, hence the displayed sequence is indeed in c_0 .

This operator is γ -summing. If $h_1, \dots, h_N$ is an orthonormal system in $L^2(0, 1)$, then the n -th coordinate of $\sum_k \gamma_k T_{M\phi} h_k$ is

$$a_n \sum_k \gamma_k \langle h_k, r_n \rangle.$$

By Bessel's inequality its variance is at most a_n^2 . Lemma therefore bounds

$$\mathbb{E} \left\| \sum_k \gamma_k T_{M\phi} h_k \right\|_{c_0}^2$$

uniformly over all finite orthonormal systems. Hence $T_{M\phi} \in \gamma_\infty(L^2(0, 1), c_0)$, as Theorem 5.2 predicts.

It remains to show that $T_{M\phi} \notin \gamma(L^2(0, 1), c_0)$. Let $R = \overline{\text{span}}\{r_n : n \geq 1\} \subseteq L^2(0, 1)$. If $T_{M\phi}$ were γ -radonifying on $L^2(0, 1)$, then its restriction to R would be γ -radonifying. Under the identification $R \simeq \ell_2$ sending r_n to the n -th unit vector, this restriction is the diagonal map

$$D : \ell_2 \rightarrow c_0, \quad D(\alpha_n) = (a_n \alpha_n)_n.$$

Thus the Gaussian series

$$\sum_{n \geq 1} \gamma_n a_n e_n$$

would converge in c_0 , and in particular the random sequence $(a_n \gamma_n)_n$ would belong to c_0 almost surely.

This is false. The events

$$A_n = \{|\gamma_n| > \sqrt{\log(n+1)}\}$$

are independent, and the standard lower Gaussian tail bound $\mathbb{P}\{|\gamma| > x\} \geq ce^{-x^2/2}/x$ for $x \geq 1$ yields

$$\sum_{n \geq 1} \mathbb{P}(A_n) \geq c \sum_{n \geq 1} \frac{1}{\sqrt{(n+1) \log(n+1)}} = \infty.$$

By Borel–Cantelli, A_n occurs infinitely often almost surely. Hence $|a_n \gamma_n| > 1$ infinitely often almost surely, so $(a_n \gamma_n)_n \notin c_0$. This contradiction proves that $T_{M\phi} \notin \gamma(L^2(0, 1), c_0)$. $\square$

Verification notes

The example satisfies the hypotheses of the source multiplier theorem: $(0, 1)$ is sigma-finite, $H = \mathbb{R}$, $E = F = c_0$, M is strongly measurable with γ -bounded range, and ϕ is constant finite rank simple. The induced operator lands in γ_∞ but not in γ , so it gives a negative answer to the proposed strengthening.

No computational verification is needed. The script `code/crop_open_problem.py` only regenerates the source crop. A bounded novelty check was performed on 2026-06-14: the run indexes were searched for arXiv:0911.3788 and core terms including γ -multiplier, γ_∞ , c_0 , and radonifying; the source section around Theorem 5.2 was inspected; web searches for the exact open-problem phrase and for gamma-multiplier/ c_0 variants found no direct answer. The source itself contains the Linde–Pietsch diagonal $\gamma_\infty \setminus \gamma$ example; the extra observation here is that Rademacher functions turn it into a multiplier with γ -bounded range.

Limitations

This packet gives a negative answer in the target space c_0 , so it is consistent with the positive statement noted in the source for target spaces not containing an isomorphic copy of c_0 . The novelty confidence is moderate until a specialist checks whether this multiplier realization is already folklore or appears implicitly elsewhere.

Human-review recommendation

Check the γ -boundedness estimate for the rank-one family $M(t) = g(t) \otimes e_1^*$, and check that restricting a γ -radonifying operator to the Rademacher subspace preserves γ -radonification. These are the two substantive steps; the rest is the standard Linde–Pietsch c_0 diagonal obstruction.

References

- [1] J. van Neerven, *γ -Radonifying operators – a survey*, arXiv:0911.3788, 2009; revised 2010.